Gnana Prakash Punnavajhala

Email | LinkedIn | Github

Education

International Institute of Information Technology, Hyderabad CGPA: 9.22/10.0
B. Tech + M.S. by Research in Computer Science and Engineering *2021 – 2026*

Research Experience

Undergraduate Researcher May 2023 – Oct 2026
Robotics Research Center *Lab Website*

- Worked on Visual Place Recognition (VPR), a complex variant of Image Retrieval, under *Prof. Madhava Krishna*
- Developed a VPR pipeline that uses cross-modal textual embeddings to overcome the performance saturation of traditional image-only architectures; under review at ICRA 2027
- Also worked on cross-modal retrieval between images and point clouds (CrossWeave, NeurIPS 2026)

AI Research Intern May 2025 – Aug 2025
IBM Research India, Gurgaon *Conversational AI Team*

- Developed *ToolWeave*, a multi-stage pipeline that synthesizes multi-turn tool-calling data by leveraging open-weight models like gpt-oss-120b to generate dialogues, overcoming the quality gaps found in non-frontier models
- Achieved over 20% performance boost for Llama-3.1-70B and Phi-4 on the *BFCL multi-turn* benchmark, outperforming existing open-source frameworks. Also demonstrated strong generalization on other benchmarks, with peak gains of 37.6% on *API Bank*, surpassing several pipelines that rely on proprietary models like GPT-4
- This work was accepted to EMNLP 2026, and two patents were filed based on it

Publications

1. Warrier, A., ***Punnavajhala, G.P.***, Tourani, S., Khan, M.H., Sharma, A., & Krishna, M. (2026). *CrossWeave: Emergent Cross-Modal Scene and Instance Retrieval from Sparse 2D-3D Alignment.* In Advances in Neural Information Processing Systems (NeurIPS 2026).
   URL https://arxiv.org/abs/2610.40525.

2. Khandelwal, D., ***Punnavajhala, G.P.***, Bhargav, G.P.S., Pandey, G., Joshi, S., Karanam, H., & Raghu, D. (2026). *ToolWeave: Structured Synthesis of Complex Multi-Turn Tool-Calling Dialogues.* In Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026).
   URL https://aclanthology.org/2026.emnlp-main.429/.

Projects

Restoring Classics to 4K | *PyTorch, NumPy, OpenCV, WandB* *Project Link*

- Implemented noise removal using Non-Local Means, Total Variation, and Wavelet Denoising, optimized via *J-invariance* (ICML 2019) for effective handling of speckle noise
- Leveraged SIFT and ResNet backbones for feature extraction on ImageNet-1k, enabling accurate chrominance prediction in YUV color space
- Enhanced colorization pipeline with joint bilateral filtering for spatial consistency and bilinear interpolation for 4K upscaling

Sinkhorn-based Aggregation for Textual Descriptors | *PyTorch, NumPy, WandB* *Project Link*

- A novel approach to enhance global descriptors of text encoders drawing inspiration from aggregation schemes used in *computer vision literature* to better capture sentence-level information
- Trained this method for the task of semantic textual similarity on datasets like MTEB, SemEval24 and Quora Question Pairs, and backbones like BERT, CLIP and RoBERTa
- Used the cosine similarity loss formulation for training, and also conducted extensive ablation studies to understand the impact of various components in the pipeline

Achievements & Honours

- Awarded the *Programme Gold Medal* for securing the highest CGPA in the Dual Degree CSE programme
- Achieved *AIR **570*** in *JEE Advanced* (top *0.38%*) and *AIR **1373*** in *JEE Mains* (top *0.13%*)
- Selected for the *Amazon ML Summer School 2024* from a pool of nearly *85,000* applicants